

to stocks, bonds, and various macro factors. The overall GSCI is a weighted combination of various commodities. With an approximate exposure of 80% to various types of oil and gas, this index heavily weights energy.

From January 1986 to December 2011, the GSCI has returned 8.9% with a standard deviation of 20.7%. The raw Sharpe ratio, without subtracting a risk-free rate, is $8.9\%/20.7\% = 0.43$. For comparison, the raw Sharpe ratios on stocks and bonds were 0.66 and 0.94, respectively, during this period. The low correlations of commodity future returns with stocks and bonds (at 14% and -9%, respectively) make commodities compelling investments. There is tremendous diversification potential in adding commodities to traditional stock and bond portfolios (see chapter 3).

Commodities offer some inflation protection, but they certainly are not a perfect hedge. The overall GSCI correlation with inflation is 29%. This reflects the relatively good inflation-hedging performance of energy futures. The correlations of crude oil and energy overall (which lumps crude oil, Brent crude oil, unleaded gas, heating oil, gas oil, and natural gas together) with inflation are 23% and 26%, respectively. Nonenergy futures are poor inflation hedgers. Agriculture, for example, has only a -4% correlation with inflation. Precious metals, which include gold, and gold on its own have low correlations of only 2% and 1%, respectively, with inflation.

Table 11.6 also lists correlations of commodities with industrial production growth (measured year on year) and the VIX volatility index. Commodity returns, especially energy and agriculture, are higher when economic growth is high. The correlations with economic growth are around 10% and so are lower than the correlations with inflation. It is notable that precious metals have slight negative correlations with growth—indicating that precious metals have some value as insurance when the economy performs poorly, but the correlations are below 10% in absolute value. Volatility is also a factor: the correlation of GSCI with VIX is -18%, which comes through energy and agricultural commodities performing poorly when volatility is high. Precious metals, in contrast, have almost no correlation with inflation.

In summary, only energy commodities have hedged inflation and even then the correlations are far from one. Energy and agricultural futures tend to perform badly when growth is low and volatility is high. Precious metals, in contrast, move fairly independently of macro factors.

5.3. SPOT, CASH, AND ROLL RETURNS

Commodity futures returns have three parts and all of them are important:

1. *Spot* return

This is the return earned by changing physical commodity prices in spot markets.